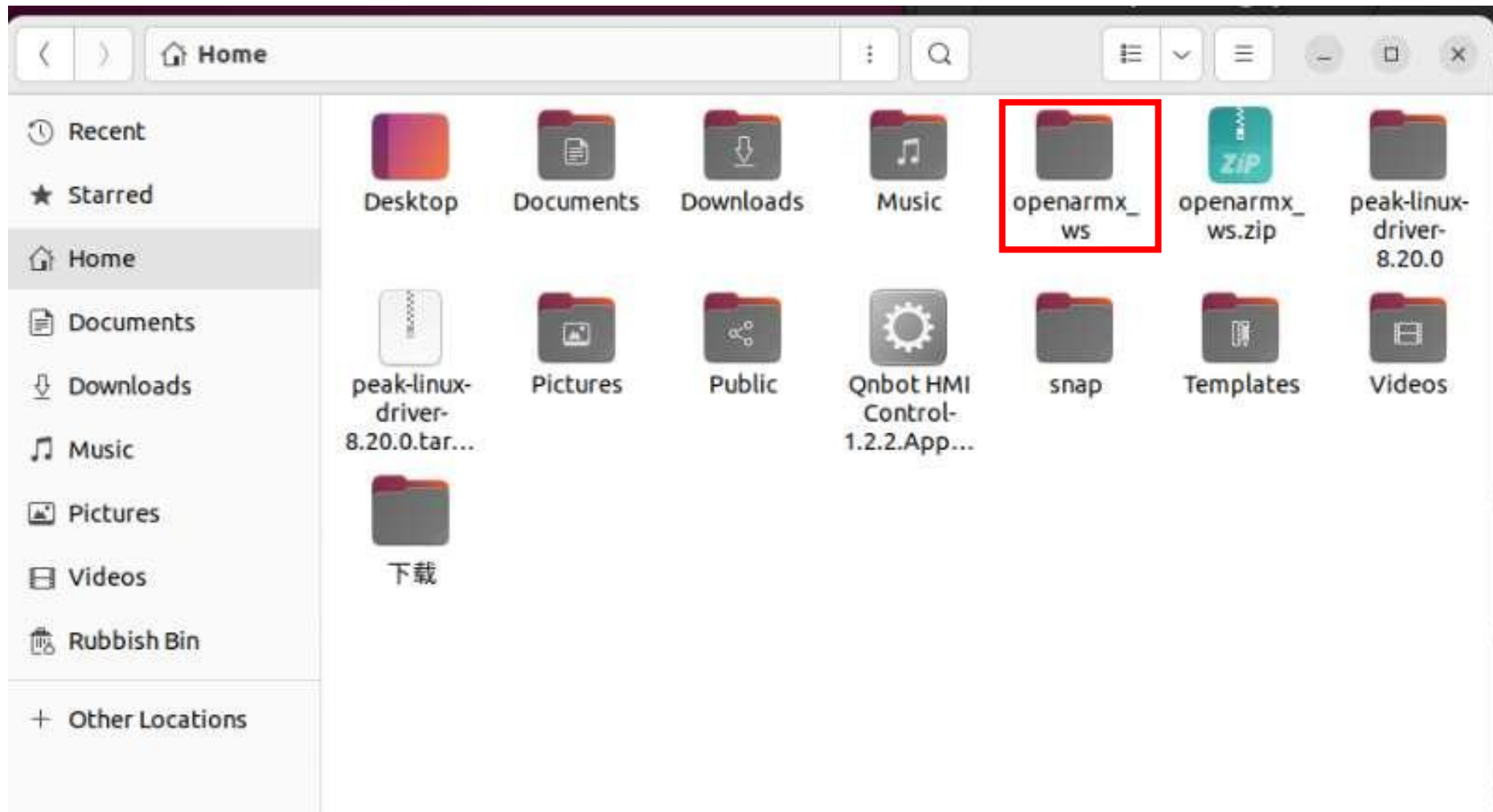
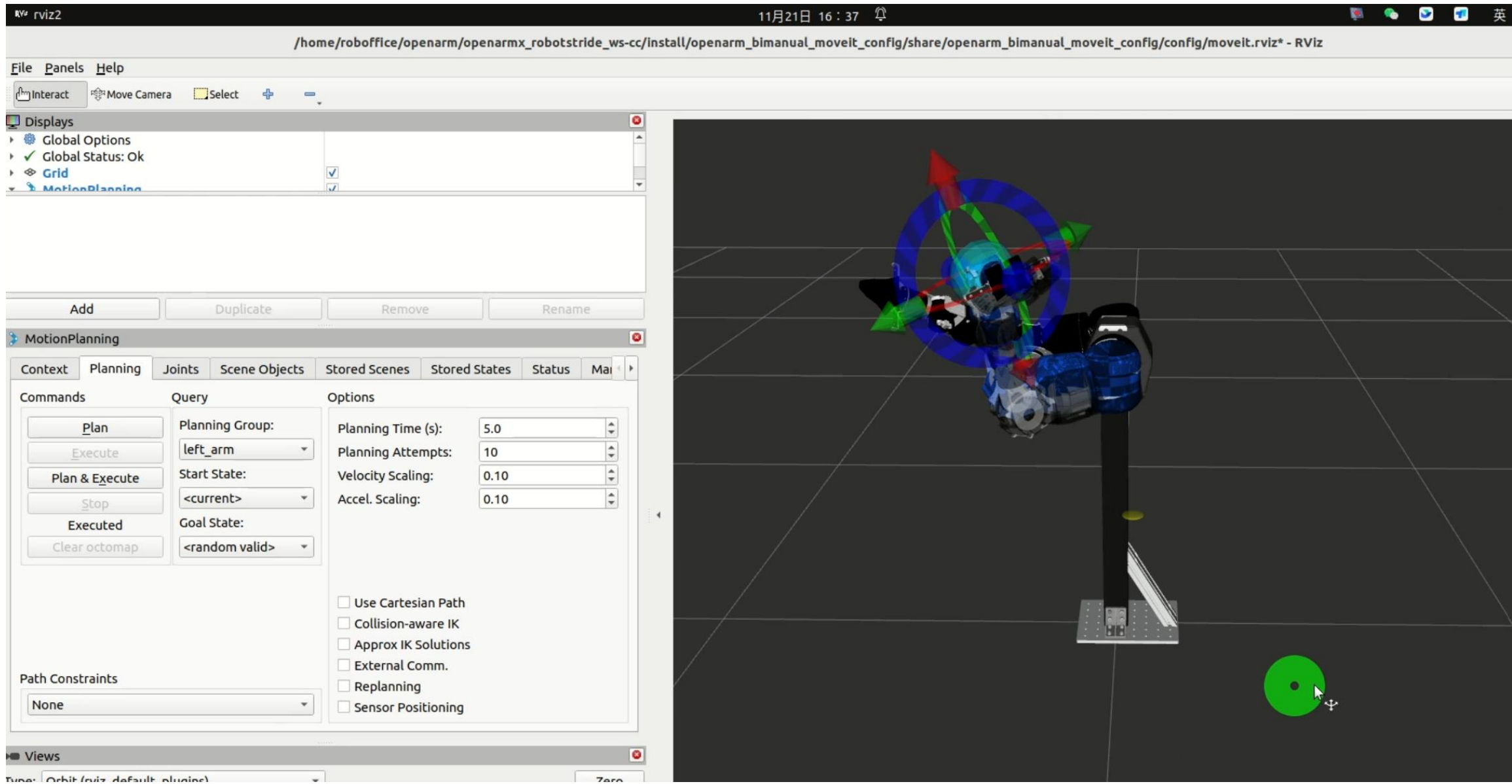




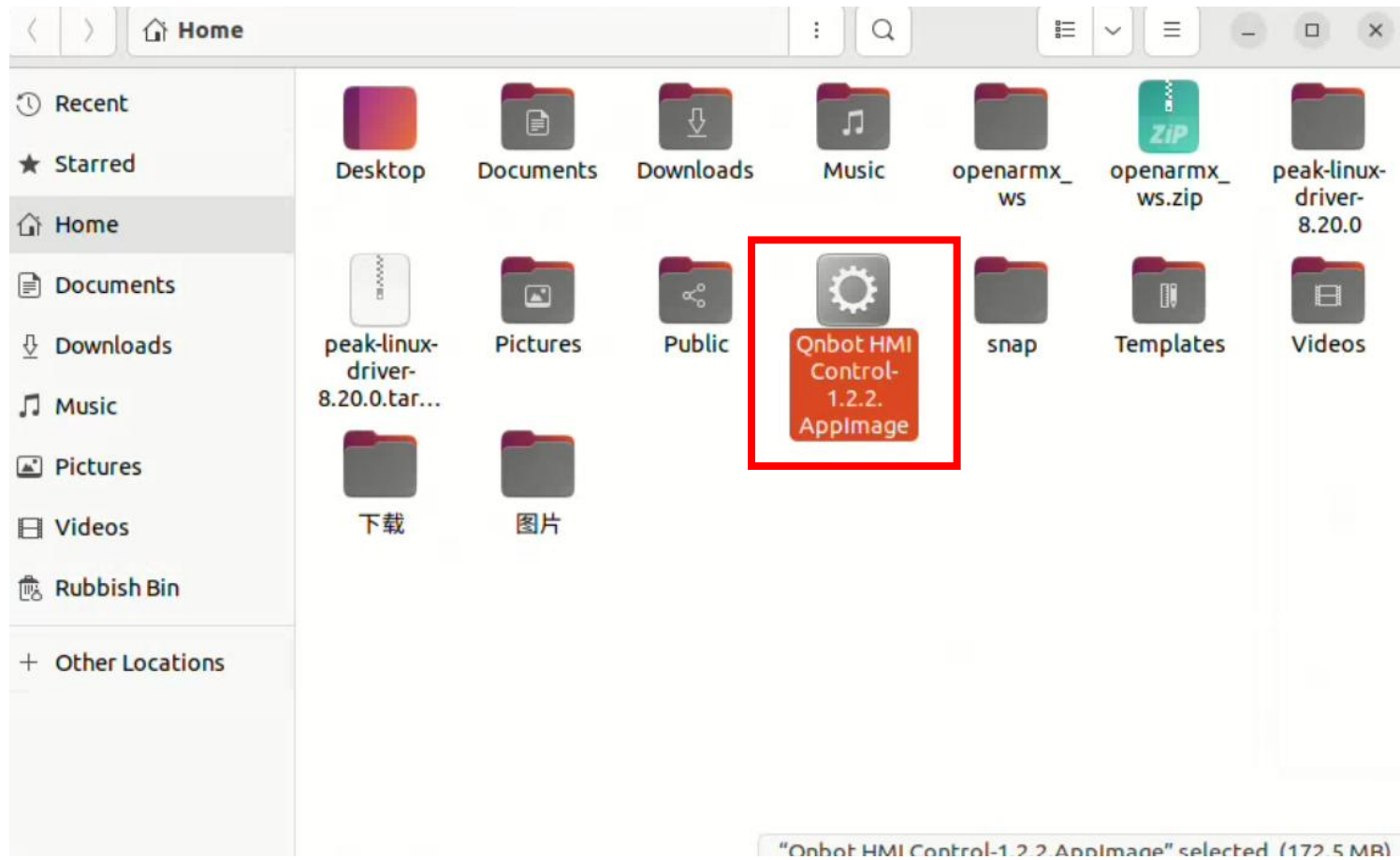
Moviet function command:

```
/home/openarmx/openarmx_ws/src/openarm_ros2/openarm_bimanual_moveit_config/run_bimanual_moveit_with_can2.0.sh
```



Exoskeleton teleoperation function:

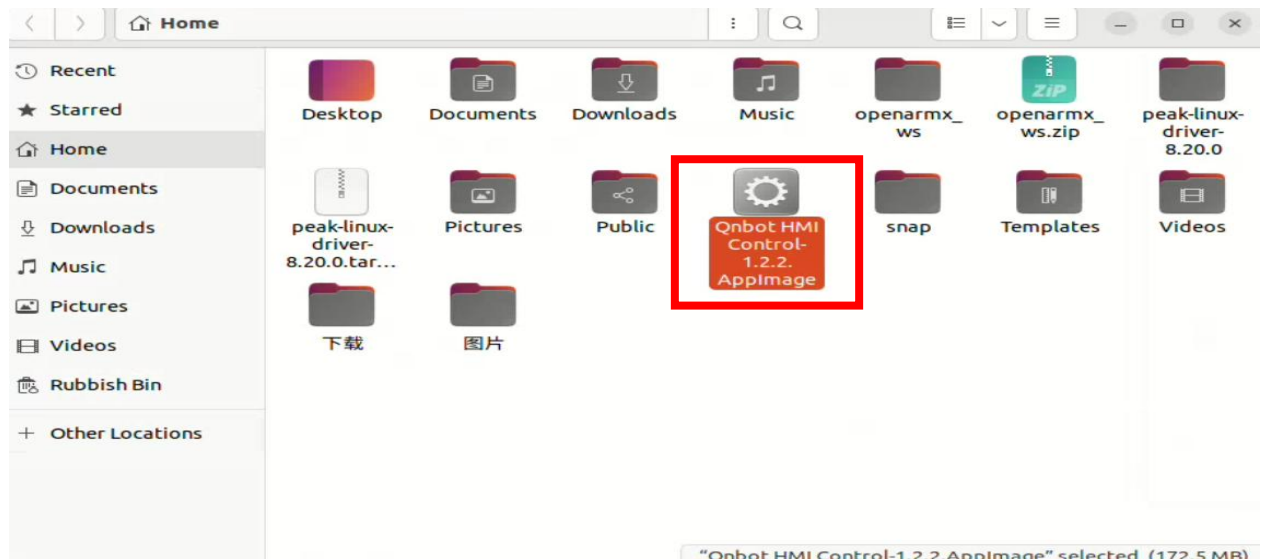
1. software



2.Connect exoskeleton teleoperation with PC:



3.Open software



Qnbot Control

Device Management

Disconnected



v1.2.2



Dashboard



Exoskeleton Device



Device Management



Recording



Data Logger



Settings

Device Control

Device Statistics

0

Total Devices

0

Connected

0

Disconnected

0

Error Status

Add New Device

Device Type

Serial

Serial Port



/dev/ttyACM0 (Qnbot Inc.) [0483:5740]

S/N: 3461345D3235

Rec



Device ID

qnbot-exoskeleton



CREATE DEVICE CONNECTION

Device List



No devices available. Please add a device connection first.



Auto-selected QNBOT device: /dev/ttyACM0



QNBOT exoskeleton controller device detected.

4.Exoskeleton teleoperation startup:

#

source install/setup.bash

(1)Start the WebSocket teleoperation service to receive exoskeleton data.

```
ros2 launch qnbot_teleoperator websocket_teleoperator.launch.py
```

(2)Start the exoskeleton redirection node to map exoskeleton data to the OpenArm robot.

```
ros2 launch qnbot_teleoperator exo_retargeting.launch.py robot_type:=OpenArm
```

(3)Add the data forwarding port address in the host computer: ws://localhost:19091.

After the successful addition, the data starts to be forwarded. Then click to stop forwarding the data.

Attention: Before the using operation, stop the data forwarding first. Wear the device correctly and ensure its proper placement to prevent the robotic arm from making sudden large - scale movements during operation, which could damage the product. You can run the simulation mode first to check the operating status of the device.

Qnbot Control

Exoskeleton Device

Connected



v1.2.2

Exoskeleton Device Data Center

+ ADD EXOSKELETON DEVICE

Configure Data Forwarding - qnbot-exoskeleton

Forwarding Target Configuration

Add New Forwarding Target

Forwarding Type

WebSocket

WebSocket URL

ws://localhost:19091

+ ADD FORWARDING TARGET



Data forwarding will send real-time raw data and parsed JSON data from exoskeleton devices to specified targets. Supports WebSocket and TCP protocols, data format is JSON.

CANCEL

APPLY CONFIGURATION

Frequency (Hz)

Data Rate (KB/s)

Total Packets

Valid Packets

548

Qnbot Control

- Dashboard
- Exoskeleton Device
- Device Management
- Recording
- Data Logger
- Settings

Exoskeleton Device

Connected



v1.2.2

Exoskeleton Device Data Center

+ ADD EXOSKELETON DEVICE

Exoskeleton Device Management



qnb-ot-exoskeleton



Status: Connected

Type: SerialAdapter

Port: /dev/ttyACM0

S/N: 3461345D3235

Data Active

Data Forwarding: 1 targets

websocket ✓

REAL-TIME STATISTICS

PARSED DATA

PARSED DATA

RAW DATA

PERFORMANCE MONITOR

qnb-ot-
exoskeletonCONFIGURE
FORWARDINGSTART
FORWARDINGSTOP
FORWARDINGREMOVE
FORWARDING

Forwarding Status

(4) Start OpenArm

(Simulation mode)

```
ros2 launch openarm_bringup openarm.bimanual.launch.py arm_type:=v10  
robot_controller:=forward_position_controller can_fd:=false use_fake_hardware:=true
```

(Real - device mode)

```
ros2 launch openarm_bringup openarm.bimanual.launch.py arm_type:=v10  
robot_controller:=forward_position_controller can_fd:=false use_fake_hardware:=false
```

Attention: Before starting the operation, it is recommended to enable the simulation mode first. Conduct actual tests only after ensuring that everything runs smoothly in the simulation.

(5) ros2 bridging control startup

```
ros2 launch qnbot_teleoperator exoskeleton_bridge.launch.py gripper_scaling_factor:=0.05
```

